

PHASE 6 – PROJECT DEVELOPMENT

6.2 PERFORMANCE TESTING

6.2.1 Introduction

Performance Testing is performed to evaluate how efficiently and reliably the developed system works under normal usage conditions. It helps identify delays, workflow issues, and other performance-related problems.

In this project, Performance Testing focuses on the ServiceNow-based automated standard laptop procurement process. The testing verifies whether requests are processed efficiently through Flow Designer and whether the automated workflow performs the required activities without unnecessary delays.

6.2.2 Objective of Performance Testing

The main objective of Performance Testing is to ensure that the automated laptop procurement workflow performs efficiently and consistently.

The objectives are:

1. To check the response of the ServiceNow request process.
2. To verify that Flow Designer triggers correctly.
3. To check the time taken for workflow activities.
4. To verify that approvals are processed correctly.
5. To ensure that tasks are created without unnecessary delay.
6. To check the reliability of status updates.
7. To verify that notifications are generated properly.
8. To identify performance issues in the automated workflow.

6.2.3 Performance Testing Process

Performance Testing starts by submitting a standard laptop request through ServiceNow. The response of the system is observed during request submission.

After the request is created, the Flow Designer workflow is monitored to verify that it starts correctly. The execution of different flow steps, such as approval processing, task creation, task assignment, and status updates, is observed.

The system is also checked to ensure that repeated requests can be processed consistently. Any unexpected delay, failed flow execution, or incorrect system response is recorded for further analysis.

6.2.4 Performance Testing Activities

Request Response Testing:

The time taken by the system to accept and create a laptop request is observed.

Flow Execution Testing:

The Flow Designer workflow is checked to ensure that it executes the required steps correctly.

Approval Processing Testing:

The approval stage is monitored to ensure that the workflow continues correctly after approval or rejection.

Task Creation Testing:

The system is checked to ensure that procurement tasks are created correctly after the required approval.

Task Assignment Testing:

The time and correctness of automatic task assignment are verified.

Status Update Testing:

The system is checked to ensure that request statuses are updated correctly during workflow execution.

Notification Testing:

Notifications are verified to ensure that they are generated correctly when important workflow events occur.

Repeated Request Testing:

Multiple requests are processed to observe whether the system continues to perform consistently.

6.2.5 Performance Factors

The following factors are considered during Performance Testing:

- System response time
- Flow execution time
- Approval processing time
- Task creation time
- Task assignment time

- Notification delivery
- Status update time
- Workflow reliability
- System stability during repeated requests

These factors help determine whether the automated process performs efficiently.

6.2.6 Performance Issues and Improvements

During testing, any delays or workflow failures are identified and analyzed. Problems may occur because of incorrect flow configuration, unnecessary workflow steps, incorrect conditions, or configuration errors.

The identified issues can be corrected by reviewing the Flow Designer configuration, simplifying unnecessary steps, checking conditions, and testing the workflow again.

Performance testing is repeated after corrections to confirm that the changes have improved the system's operation.

6.2.7 Expected Outcome

The expected outcome of Performance Testing is a reliable and efficient automated procurement workflow. The system should process standard laptop requests smoothly and execute the required Flow Designer activities without unnecessary delays.

The workflow should consistently perform request processing, approval, task creation, assignment, status updates, and notifications.

6.2.8 Conclusion

Performance Testing ensures that the automated standard laptop procurement system operates efficiently under expected usage conditions. By testing the response and execution of the ServiceNow workflow, performance issues can be identified and corrected before the final project implementation.

This testing helps ensure that Flow Designer provides a stable and dependable automation process for standard laptop procurement.